

TAXI PAYMENT TYPE REVENUE ANALYSIS

Payment Type & Fare Analysis

Transactions Analyzed	Card Payments	Cash Payments
6.4M+	67.5%	32.5%

1. Business Problem

Taxi drivers need to understand whether payment methods are associated with different fare amounts and customer trip characteristics in order to identify potential opportunities to improve revenue.

2. Analysis Objective

Analyze taxi trip data to compare card and cash transactions, identify differences in fare amount and trip characteristics, and evaluate whether the observed fare difference between payment methods is statistically significant.

3. Dataset Overview

- 6.4M+ NYC taxi trip records.
- Analysis focused on passenger count, payment type, fare amount, trip distance, and trip duration.
- Card and cash transactions were used for the primary comparison.
- Records with invalid passenger counts, non-positive fares, distances, or durations were excluded from the analysis.

4. Key Findings

Payment Type & Fare

- Card transactions represent approximately 67.5% of analyzed transactions, compared with 32.5% for cash.
- Card trips show a higher average fare amount than cash trips.
- Card trips also show a slightly higher average trip distance than cash trips.

Passenger & Trip Patterns

- Single-passenger trips represent the largest share of both card and cash transactions.
- The proportion of transactions generally decreases as passenger count increases.
- The observed fare difference should be considered alongside trip distance and duration because these characteristics also differ between payment groups.

5. Hypothesis Testing

- Null Hypothesis (H_0): There is no difference in average fare between card and cash transactions.
- Alternative Hypothesis (H_1): There is a difference in average fare between card and cash transactions.

- Welch's independent two-sample t-test was used to compare the average fare amounts.
- The test produced a t-statistic of approximately 170.38 with a p-value effectively equal to 0, indicating a statistically significant difference in observed average fares.

6. Business Implications

Card payments are associated with higher observed average fares and longer average trip distances in the analyzed data. This identifies payment type as a useful business variable for further revenue analysis.

7. Recommended Areas for Further Investigation

- Investigate whether the fare difference remains after accounting for trip distance and duration.
- Compare fare and trip characteristics across different time periods and trip segments.
- Evaluate customer payment behavior before introducing payment incentives.
- Assess total trip amount and tip behavior to provide a broader view of transaction value.

8. Conclusion

The analysis shows a statistically significant difference in observed average fares between card and cash transactions, with card transactions associated with higher average fares in this dataset. Further segmented analysis is recommended before using payment type as a direct revenue-driving strategy.